

The finite J -norm is dominated by the discrete K -norm
(the direction $\mathcal{J} \lesssim \|\cdot\|_K$ of Janson's Lemma 1, over a monoid)

1 Setting and statement

Let α be an additive commutative monoid with a preorder $\leq$ and a map $x \mapsto |x|$ from α to itself satisfying

$$|x + y| \leq |x| + |y| \quad (x, y \in \alpha), \quad (1)$$

together with the following *refined Riesz decomposition*:

(R) if $|x| \leq |y| + |z|$, then there are $x_0, x_1 \in \alpha$ with

$$x = x_0 + x_1, \quad |x_0| \leq |y|, \quad |x_1| \leq |z|, \quad |x_0| \leq |x|, \quad |x_1| \leq |x|.$$

The first three conditions of (R) are the field `exists_decomp` of `Abs.IsModulus`; the last two are new. They hold for the three existing instances with the witnesses already used there: on $\mathbb{R}$, x_0 is x clamped to $[-|y|, |y|]$ and $x_1 = x - x_0$ has the sign of x ; on $[0, \infty]$, $x_0 = \min(x, y)$ and $x_1 = x - x_0$; on products, (R) is checked pointwise.

Let A_0, A_1 be quasinorms on α with values in $[0, \infty]$ that are *solid*: $|x| \leq |y|$ implies $\|x\|_i \leq \|y\|_i$, where $\|\cdot\|_i$ is the quasinorm of A_i . No other property of A_0, A_1 is used; in particular their quasi-triangle constants never appear.

Fix $0 < \theta < 1$ and $0 < q \leq \infty$. For $\nu \in \mathbb{Z}$, $x, u \in \alpha$ put

$$K_\nu(x) = \inf_{x=x_0+x_1} (\|x_0\|_0 + 2^\nu \|x_1\|_1), \quad J_\nu(u) = \max(\|u\|_0, 2^\nu \|u\|_1),$$

and $\Delta = \{x \in \alpha : \|x\|_0 < \infty, \|x\|_1 < \infty\}$. For a family $f = (f_\nu)_{\nu \in S}$ in $[0, \infty]$ indexed by a countable set S , $\|f\|_{\ell^q}$ is $(\sum_{\nu \in S} f_\nu^q)^{1/q}$ if $q < \infty$ and $\sup_{\nu \in S} f_\nu$ if $q = \infty$. The two norms to be compared are

$$\|x\|_K = \|(2^{-\nu\theta} K_\nu(x))_{\nu \in \mathbb{Z}}\|_{\ell^q} \quad (\text{discreteKMethod}),$$

$$\mathcal{J}(x) = \inf \left\{ \|(2^{-\nu\theta} J_\nu(u_\nu))_{\nu \in W_N}\|_{\ell^q} : N \in \mathbb{N}, u : W_N \rightarrow \alpha, \sum_{\nu \in W_N} u_\nu = x \right\} \quad (\text{jInfNorm}),$$

where $W_N = \{-N, \dots, N-1\}$. In Lean the window is indexed by $k \in \text{Fin}(2N)$ with $\nu = k - N$.

Theorem 1. *For every $x \in \Delta$,*

$$\mathcal{J}(x) \leq 2^{\min(\theta, 1-\theta)} \|x\|_K.$$

The constant is at most $\sqrt{2}$ and does not depend on q or on the couple.

Idea of the proof. For each level μ choose a near-optimal decomposition $x = y_\mu + z_\mu$ for $K_\mu(x)$. Using (R), split x into $2N$ pieces, the piece u_ν being dominated by $y_{\nu+1}$ and by z_ν . Then $J_\nu(u_\nu) \lesssim K_{\nu+1}(x)$, except for one of the two norms of each of the two end pieces, which is controlled only by x itself. After weighting, these end terms decay geometrically in N , because $0 < \theta < 1$ and $x \in \Delta$. The ℓ^q -norm of a maximum is estimated by a sum of q -th powers, so no quasi-triangle inequality in ℓ^q is needed.

2 Three elementary facts

Lemma 2. *For every $x \in \alpha$ and $\nu \in \mathbb{Z}$, $K_\nu(x) \leq K_{\nu+1}(x) \leq 2K_\nu(x)$.*

Proof. For any decomposition $x = x_0 + x_1$,

$$\|x_0\|_0 + 2^\nu \|x_1\|_1 \leq \|x_0\|_0 + 2^{\nu+1} \|x_1\|_1 \leq 2(\|x_0\|_0 + 2^\nu \|x_1\|_1).$$

Taking the infimum over all decompositions gives both inequalities. $\square$

No finiteness assumption on x is needed; the hypothesis of `monotone_kNorm` in `Basic.lean` is superfluous for dyadic points. The second inequality is a case of `kNorm_le_mul`. The main proof uses only the first one.

Lemma 3. *For every $x \in \alpha$, $\mu \in \mathbb{Z}$ and $\delta > 0$ there is a decomposition $x = y + z$ with $\|y\|_0 + 2^\mu \|z\|_1 \leq K_\mu(x) + \delta$.*

Proof. If $K_\mu(x) = \infty$, any decomposition works, e.g. $x = 0 + x$. Otherwise $K_\mu(x) < \infty$, and by the definition of the infimum some decomposition has cost below $K_\mu(x) + \delta$ (`exists_decomp_lt_of_lt_kNorm`). $\square$

An additive error is used because the infimum need not be attained, and a multiplicative error $1 + \varepsilon$ is useless when $K_\mu(x) = 0$.

Lemma 4. *Let f, g be families in $[0, \infty]$ indexed by a countable set S . If $q < \infty$, then $\|\max(f, g)\|_{\ell^q}^q \leq \|f\|_{\ell^q}^q + \|g\|_{\ell^q}^q$. If $q = \infty$, then $\|\max(f, g)\|_{\ell^\infty} = \max(\|f\|_{\ell^\infty}, \|g\|_{\ell^\infty})$.*

Proof. For $q < \infty$, sum the pointwise inequality $\max(s, t)^q \leq s^q + t^q$ over S . For $q = \infty$, both sides are the supremum of all values of f and g . $\square$

3 Splitting x into $2N$ pieces

This is the only place where (R) is used. The pieces are split off one at a time from the part of x not yet assigned, so no subtraction is needed.

Proposition 5. *Let $N \geq 1$, $x \in \alpha$, and for $\mu = -N + 1, \dots, N - 1$ let $x = y_\mu + z_\mu$ be arbitrary decompositions. Then there are $u_\nu \in \alpha$, $\nu \in W_N$, with $\sum_{\nu \in W_N} u_\nu = x$ and, for every $\nu \in W_N$,*

- (a) $|u_\nu| \leq |y_{\nu+1}|$ if $\nu \leq N - 2$;
- (b) $|u_\nu| \leq |z_\nu|$ if $\nu \geq -N + 1$;
- (c) $|u_\nu| \leq |x|$.

Proof. We construct remainders $r_{-N}, \dots, r_{N-1}$ and pieces $u_{-N}, \dots, u_{N-2}$ recursively, maintaining for each ν the invariant

$$x = \sum_{-N \leq \kappa < \nu} u_\kappa + r_\nu, \quad |r_\nu| \leq |x|. \quad (2)$$

Start with $r_{-N} = x$; then (2) holds by reflexivity, the sum being empty.

Recursive step. Let $-N \leq \nu \leq N-2$ and suppose r_ν satisfies (2). Since $x = y_{\nu+1} + z_{\nu+1}$, inequality (1) and transitivity give

$$|r_\nu| \leq |x| \leq |y_{\nu+1}| + |z_{\nu+1}|.$$

Apply (R) to r_ν . It gives $r_\nu = u_\nu + r_{\nu+1}$ with

$$|u_\nu| \leq |y_{\nu+1}|, \quad |r_{\nu+1}| \leq |z_{\nu+1}|, \quad |u_\nu| \leq |r_\nu|, \quad |r_{\nu+1}| \leq |r_\nu|. \quad (3)$$

Substituting $r_\nu = u_\nu + r_{\nu+1}$ into (2) and using associativity gives the identity in (2) for $\nu+1$. The last inequality in (3) and $|r_\nu| \leq |x|$ give $|r_{\nu+1}| \leq |x|$.

Last piece. Put $u_{N-1} = r_{N-1}$. The invariant (2) for $\nu = N-1$ reads $x = \sum_{\nu \in W_N} u_\nu$.

The bounds. Property (a) is the first inequality of (3). For (b) and (c) note that $|u_\nu| \leq |r_\nu|$ for all $\nu \in W_N$: for $\nu \leq N-2$ by (3), and for $\nu = N-1$ by definition. Now $|r_\nu| \leq |x|$ by (2), which gives (c); and for $\nu \geq -N+1$, $|r_\nu| \leq |z_\nu|$ by the second inequality of (3) at step $\nu-1$, which gives (b). $\square$

Both new conditions of (R) are needed. The condition $|x_1| \leq |x|$ keeps the remainders below $|x|$, without which (R) could not be applied at the next step; the condition $|x_0| \leq |x|$ passes the domination of the remainder by z_ν to the new piece.

4 The pointwise estimate

Lemma 6. *Let $x \in \alpha$, $\delta > 0$ and $N \geq 1$. For $\mu = -N+1, \dots, N-1$ choose decompositions $x = y_\mu + z_\mu$ as in Lemma 3, i.e.*

$$\|y_\mu\|_0 + 2^\mu \|z_\mu\|_1 \leq K_\mu(x) + \delta, \quad (4)$$

and let $(u_\nu)_{\nu \in W_N}$ be given by Proposition 5. Put $E_{-N} = 2^{-N}\|x\|_1$, $E_{N-1} = \|x\|_0$ and $E_\nu = 0$ for the other $\nu \in W_N$. Then for every $\nu \in W_N$,

$$J_\nu(u_\nu) \leq \max(K_{\nu+1}(x) + \delta, E_\nu).$$

Proof. Since $N \geq 1$, the indices $-N$ and $N-1$ are distinct, so each ν is exceptional in at most one of the two entries of $J_\nu(u_\nu)$. We bound both entries by the right-hand side; solidity turns each modulus inequality of Proposition 5 into an inequality of quasinorms.

The entry $\|u_\nu\|_0$. If $\nu \leq N-2$, then by (a) and (4) at $\mu = \nu+1$, $\|u_\nu\|_0 \leq \|y_{\nu+1}\|_0 \leq K_{\nu+1}(x) + \delta$. If $\nu = N-1$, then by (c), $\|u_\nu\|_0 \leq \|x\|_0 = E_{N-1}$.

The entry $2^\nu \|u_\nu\|_1$. If $\nu \geq -N+1$, then by (b), (4) at $\mu = \nu$, and Lemma 2, $2^\nu \|u_\nu\|_1 \leq 2^\nu \|z_\nu\|_1 \leq K_\nu(x) + \delta \leq K_{\nu+1}(x) + \delta$. If $\nu = -N$, then by (c), $2^{-N} \|u_\nu\|_1 \leq 2^{-N} \|x\|_1 = E_{-N}$. $\square$

5 Proof of Theorem 1

We prove $\mathcal{J}(x) \leq 2^\theta \|x\|_K$; the bound with $2^{1-\theta}$ is Remark 7. Fix $x \in \Delta$, $N \geq 1$ and $\delta > 0$, and let u be as in Lemma 6. Write

$$a_\mu = 2^{-\mu\theta} K_\mu(x), \quad b_\nu = 2^{-\nu\theta} J_\nu(u_\nu), \quad e_\nu = 2^{-\nu\theta} E_\nu.$$

Since $2^{-\nu\theta} = 2^\theta \cdot 2^{-(\nu+1)\theta}$, multiplying Lemma 6 by $2^{-\nu\theta}$ gives

$$b_\nu \leq \max(2^\theta (a_{\nu+1} + \delta 2^{-(\nu+1)\theta}), e_\nu) \quad (\nu \in W_N). \quad (5)$$

The end terms are

$$e_{-N} = 2^{N\theta} 2^{-N} \|x\|_1 = 2^{-(1-\theta)N} \|x\|_1, \quad e_{N-1} = 2^{-(N-1)\theta} \|x\|_0.$$

Since $\sum_\nu u_\nu = x$, the family u is admissible in the definition of $\mathcal{J}$, so $\mathcal{J}(x) \leq \|b\|_{\ell^q}$.

Case $q < \infty$. By (5), Lemma 4, and the fact that e vanishes except at $-N$ and $N-1$,

$$\mathcal{J}(x)^q \leq \sum_{\nu \in W_N} b_\nu^q \leq 2^{\theta q} \sum_{\nu \in W_N} (a_{\nu+1} + \delta 2^{-(\nu+1)\theta})^q + e_{-N}^q + e_{N-1}^q.$$

The left-hand side does not depend on δ . The right-hand side is a finite sum of functions of δ that are continuous on $[0, \infty)$ with values in $[0, \infty]$ (addition, multiplication by a finite constant, and $t \mapsto t^q$ are continuous there). Letting $\delta \rightarrow 0$ and substituting $\mu = \nu + 1$,

$$\mathcal{J}(x)^q \leq 2^{\theta q} \sum_{\mu=-N+1}^N a_\mu^q + e_{-N}^q + e_{N-1}^q \leq 2^{\theta q} \|x\|_K^q + e_{-N}^q + e_{N-1}^q,$$

where the last step drops the terms with $\mu \notin \{-N+1, \dots, N\}$. Since $0 < \theta < 1$ and $\|x\|_0, \|x\|_1 < \infty$, both e_{-N} and e_{N-1} tend to 0 as $N \rightarrow \infty$, and so do their q -th powers. Letting $N \rightarrow \infty$ gives $\mathcal{J}(x)^q \leq 2^{\theta q} \|x\|_K^q$, and taking q -th roots proves the claim.

Case $q = \infty$. By (5) and Lemma 4, and since $2^{-(\nu+1)\theta} \leq 2^{(N-1)\theta}$ on W_N ,

$$\mathcal{J}(x) \leq \max_{\nu \in W_N} b_\nu \leq \max(2^\theta (\|x\|_K + \delta 2^{(N-1)\theta}), e_{-N}, e_{N-1}).$$

Let $\delta \rightarrow 0$, then $N \rightarrow \infty$. □

The two limits are taken one after the other: $\delta \rightarrow 0$ for fixed N , then $N \rightarrow \infty$. This is legitimate because $\mathcal{J}(x)$ depends on neither, and the end terms do not depend on δ . No assumption $\|x\|_K < \infty$ is needed.

6 Remarks

Remark 7 (The constant $2^{1-\theta}$). Feed Proposition 5 the decompositions one level lower: for $\mu = -N+1, \dots, N-1$, let $x = y_\mu + z_\mu$ satisfy (4) at level $\mu-1$. For $\nu \leq N-2$, (a) and Lemma 2 give $\|u_\nu\|_0 \leq K_\nu(x) + \delta \leq 2(K_{\nu-1}(x) + \delta)$; for $\nu \geq -N+1$, (b) gives $2^\nu \|u_\nu\|_1 = 2 \cdot 2^{\nu-1} \|u_\nu\|_1 \leq 2(K_{\nu-1}(x) + \delta)$. The exceptional entries are unchanged, so $J_\nu(u_\nu) \leq \max(2(K_{\nu-1}(x) + \delta), E_\nu)$. Since $2 \cdot 2^{-\nu\theta} = 2^{1-\theta} 2^{-(\nu-1)\theta}$, the rest of the proof goes through with $2^\theta a_{\nu+1}$ replaced by $2^{1-\theta} a_{\nu-1}$. This uses the second inequality of Lemma 2. Choosing the better of the two versions gives the constant $2^{\min(\theta, 1-\theta)}$.

Remark 8 (Where the hypotheses are used).

- (R), (1) and transitivity of $\leq$: only in Proposition 5.
- Solidity of A_0, A_1 : only in Lemma 6.
- $x \in \Delta$ and $0 < \theta < 1$: only in $e_{-N}, e_{N-1} \rightarrow 0$. Each end uses one half of each condition: e_{-N} uses $\theta < 1$ and $\|x\|_1 < \infty$, e_{N-1} uses $\theta > 0$ and $\|x\|_0 < \infty$.
- The quasi-triangle inequalities of A_0, A_1 and of ℓ^q : never. Lemma 4 replaces Minkowski's inequality, which is why the constant does not depend on q .

For $q = 0$, Mathlib's `eLpNorm` is identically 0, so $\mathcal{J} \equiv 0$ and the inequality is trivial.

Remark 9 (Why $x \in \Delta$ is necessary). Let $q \neq 0$. If $\mathcal{J}(x) < \infty$, some admissible ℓ^q -norm is finite, so each $J_\nu(u_\nu)$ is finite and each u_ν lies in Δ . Since Δ is an additive submonoid (by the quasi-triangle inequality with finite constants), the finite sum x lies in Δ . Hence $\mathcal{J} = \infty$ outside Δ , whereas $\|x\|_K$ may be finite there.